Atharva Khambete

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Education

Marathwada Mitra Mandal's College of Engineering (MMCOE), SPPU Expected June 2027
B.E. in Information Technology | CGPA: 9.5/10 *Pune, India*

Experience

Qutrit Innovations Pvt. Ltd. Pune, India
Cloud and Security Intern *Feb 2026 – Present*

- Engineered the SchemaMatrix Delta Engine (SMDE) using a stateless, baseline-default architecture, eliminating the computational overhead and data corruption risks caused by dropped packets in traditional stateful delta encoding.
- Reduced IoT telemetry payload size by 21× relative to raw JSON by developing SBDBC-TS, a custom embedded storage engine featuring an append-only write pipeline, bitmask-driven sparse storage, and delta encoding.
- Deployed light-weight Edge AI models for real-time anomaly detection on streaming sensor telemetry, optimizing edge-side preprocessing to reduce cloud data transmission overhead.

Research Intern *Aug 2025 – Oct 2025*

- Designed edge-to-cloud IoT architectures for telemetry collection and distributed sensor communication systems.
- Reduced operational costs by 50% by leveraging AWS cloud computing principles, including spot instances and optimized lifecycle policies.

Projects

NATO: Novel Adaptive Training Optimizer Patent Published 2025
application number : 202521103141

- Designed a deep-learning optimizer that efficiently guides training toward flat, noise-resilient minima to improve generalization, calibration, and adversarial robustness without the large computational overhead of methods such as Sharpness-Aware Minimization.
- Bench-marked optimizer behavior against SGD and Adam across wide range of tasks, achieving a p-value < 0.001 for CIFAR10, CIFAR100, SVHN, and DESI.
- Published official PyPI package with over 2.2k downloads; metrics available here.

Security Feedback Control Architecture (SFCA) IEEE Cloud Summit 2026, Washington DC
View Session Link

- Designed the Security Feedback Control Architecture to enforce Input-to-State Stability in enterprise infrastructures, preventing sequential defensive actions from causing service disruption.
- Built a 3-layered framework for automated security response coordination.

- Utilized Model Predictive Control for handling multi-event attack scenarios.
- Reduced system violations by 9-10x compared to state-of-the-art baselines.

Secure-Torch: Defense-in-Depth Trust Gateway for AI Runtime Security — IJIRT Publication

View Paper

- Published a Python library and research paper in the International Journal of Innovative Research in Technology (IJIRT) addressing AI model runtime security vulnerabilities.
- Developed a defense-in-depth trust gateway for secure AI model deserialization, actively preventing arbitrary code execution during model loading phases.
- Maintained the official PyPI release with over 1.6k+ downloads; package metrics trackable here.

Lamarr-Turing Graphs: DRL-based Adaptive Network Routing — IEEE ICFST 2026

- Developed a reinforcement learning-based adaptive routing framework to overcome the lack of intelligent adaptivity in static networks, ensuring resilient communication during active adversarial disruptions.
- Evaluated routing behavior under adversarial simulation scenarios using OMNET++.
- Improved Packet Delivery Ratio (PDR) by [Insert %] under simulated adversarial environments.

Technical Skills

Languages
Python, C++, C, Rust, Java, Bash

Cloud and Infrastructure
AWS, Docker, Kubernetes, Linux, MQTT, Terraform

Machine Learning
PyTorch, TensorFlow, Scikit-learn, Deep Learning, Reinforcement Learning

Security and Systems
TLS, X.509, eBPF, IoT Systems, Network Security

Tools
Git, GitHub, Docker Compose, OMNET++